

400 Hz Crystal-Controlled Sine Wave Inverters

Precision AC Power for Aerospace, Defense & Mission-Critical Applications

- Stable Pure Sine Output.
- High-Power AC Conversion.
- Built for Demanding Applications.



CRYONANO 400 Hz Crystal-Controlled Sine Wave Inverters are engineered to deliver highly stable, low-distortion AC power for applications requiring exceptional waveform purity, frequency accuracy, and operational reliability. Designed for both single-phase and three-phase systems, these inverters convert standard AC input into precision-controlled 400 Hz output power for aerospace ground support, avionics testing, naval systems, defense platforms, and advanced industrial environments. Built using industrial-grade power electronics and a rugged modular architecture, the systems maintain stable operation under complex, dynamic, and unbalanced load conditions.

KEY FEATURES

- Precision Frequency Control**
 Crystal-controlled output frequency with fixed options of 50 Hz, 60 Hz, 400 Hz, and 800 Hz, plus programmable ranges from 40–400 Hz and 400–800 Hz for high stability and accuracy.
- Pure Sine Wave Output**
 Low-distortion pure sine wave architecture maintains waveform integrity under varying loads for avionics, radar electronics, flight-control systems, and precision instrumentation.
- Scalable Power Architecture**
 Modular system available from 500 VA to 45 kVA supports scalable deployment across single-phase and three-phase configurations.
- Flexible Voltage Configurations**
 Supports multiple input/output voltage configurations including 115 V AC, 230 V AC, 200 V AC, 400 V AC, and 480 V AC, with custom voltage options available.

Notes

- Output configuration and cooling depend on power level and deployment requirements
- Custom frequency and voltage configurations available
- Designed for operation in high-reliability environments
- Suitable for both laboratory and field deployment

APPLICATIONS

- Aerospace ground support equipment
- Avionics and flight-control testing
- Naval and shipboard systems
- Radar and defense electronics
- Industrial frequency conversion systems
- Laboratory and precision instrumentation testing



DC-AC Pure Sine Wave Inverters

SPECIFICATIONS

ELECTRICAL PERFORMANCE

Power Range	500 VA – 45 kVA
Output Frequency	50 / 60 / 400 / 800 Hz
Programmable Frequency	Range 40–800 Hz
Output Waveform	Pure sine wave
Input Voltage	115–480 V AC
Output Voltage	Model dependent / configurable

LOAD & SYSTEM COMPATIBILITY

Load Type	Balanced & unbalanced loads
Surge Capability	High inrush load support
Frequency Stability	Crystal-controlled
Output Quality	Low distortion

PROTECTION & SAFETY

Short-Circuit Protection	Continuous
Thermal Protection	Integrated shutdown
Overload Protection	Integrated
Automatic Restart	Supported

MECHANICAL & SYSTEM DESIGN

Architecture	Modular, scalable
Construction	Industrial-grade rugged design
Power Density	High-density configuration
Cooling	Configuration dependent

CONFIGURABILITY

Architecture Options	COTS / Modified / Fully Custom
Form Factor	Embedded / rack / standalone
Output Configuration	Single / multi-channel
Customization	Voltage, current, environment-specific

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